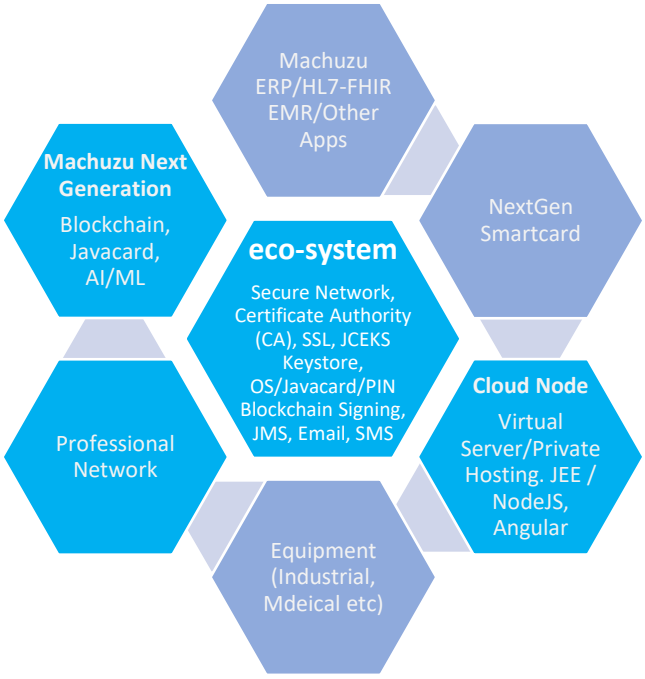
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Machuzu Eco-System

07 September 2026



To be operating within the eco-system, below are the minimum applications that you have to be running or subscribed to;

1. Machuzu Next Generation

This is a JavaFX Desktop application. It stores data in a local immutable database ImmuDB and other data in SQLite. The app performs rapid processing of complex operations including Blockchain and AI. This app can be compiled to a custom mobile app at a premium.

2. Cloud Node

Considering no Peer-To-Peer network exists; this is the most affordable method to have a secure data sharing network for Blockchain. This is a cloud hosted communications Node; it receives and stores hashed data essential to sharing across the network.

3. Professional Network

You are only operating within the *eco-system* if you share data within the *decentralised network*, among other participating nodes which you can invite or be invited by.

4. Costs

Applications for a quotation need to be submitted. It must be demonstrated that a professional network exists. This is relevant for the above-mentioned products and services only.

Additional Components

In addition to the standard or mandatory items; the below can be implemented outside of the eco-system. They integrate closely and are within the eco-system if used with the mandatory items.

1. *Machuzu ERP/HL7-FHIR EMR*
2. *NextGen Smartcard*
3. *MaxYield (Farm-To-Fork/Sovereign Trace)*
4. *Umsiti Operational Control Software*
5. *Equipment (Industrial, Medical etc)*

Minimum Requirements For Each's Use

1. Machuzu ERP/HL7-FHIR EMR
   Desktop or Server with at least 2 GB RAM
2. NextGen Smartcard
   Smartcard Reader, Compatible Smartcard, Machuzu ERP/HL7-FHIR EMR, NextGen Blockchain / Sovereign Trace, Eco-System
1. Machuzu NextGen App (Blockchain Identity & Sovereign Trace)
   Laptop/Desktop/Server with 4 GB RAM
3. MaxYield (Farm-To-Fork/Sovereign Trace)
   Eco-System, Cloud Node
4. Machuzu Secure Network & Cloud Node
   Eco-System

Function Individually or within the Eco-System

2. Machuzu ERP/HL7-FHIR EMR

Open Source Core Code. Premium Modules, Customisation and Consulting.

Key Features

Machuzu ERP: Accounting, CRM, SCM, HRM, POS. Project Management, Collaboration / Document Management, Custom Workflow.

HL7-FHIR EMR: Appointments, Vitals, Allergy, Condition, Procedure, Diagnosis, Medication, Lab, Observation, Immunisation, Document Management, Financial. Public and Private Facility workflow and specification customisation.

Technology

Jakarta EE (Java EE), Wildfly 24, Javaserver Faces (JSF), Primefaces, MySQL

What it can be used for

- To securely capture all operational data and monitor strategic objectives. Contains basic reports and dashboards. Refer to specific data about products for more information.

Integrations

- Share data within a secure custom network or regular API's, with similar or different ERP systems.
- Link Wildfly Credential Manager to Smartcard storage and Next Generation Blockchain Identity.
- Smartcard Signing of transactions

3. NextGen Smartcard

Standard Encryption. Premium services include: On-Chip Biometric Storage, Post Quantum Cryptography Encryption. Private Key Generation for non-eco-system blockchain based applications. Off-the-shelf products include Smartcard Reader and Smartcards.

Technology

Javacard, GlobalPlatform

What it can be used for

- Allows organisations and individuals to securely store their Blockchain Identities or Machuzu ERP/HL7-FHIR EMR IPA Profile

Integrations

- Can be used for authentication, approvals or sign-offs in any relevant application. See file NextGen/ MachuzuNextGen.pdf

4. Machuzu NextGen App (Blockchain Identity & Sovereign Trace)

Standard Encryption. Certificate Authority. Java JCEKS Keystore. Free Download. Premium services include: Cloud Node Hosting and Management. Peer-To-Peer Network Configuration and Administration. Custom Blockchain. AI Model Training.

Technology

JavaFx, Blockchain, ImmuDb, SQLite, Gluon, AI (Deeplearning4j, LangChain4j)

What it can be used for

- To securely store Identifying and Sovereign Trace data
- To enable decentralised storage (backup) of data within a secure custom network
- AI Model Creation and Training
- Reporting and Analytics Front-End

- Smartcard Management (Write and Manage Data on Card)

Integrations

- Can be used to login to Machuzu ERP/HL7-FHIR EMR and other systems, apps or email automatically or at a click of a button; keeping a Blockchain Audit Trail
- Operating System (Windows, Mac, Linux) Level Signing

5. MaxYield (Farm-To-Fork/Sovereign Trace)

Standard Encryption. Free Subscription. Premium services include: Cloud Hosting and Management. Machuzu NextGen App.

What it can be used for

- To track crop Planting to Harvesting

Integrations

- Machuzu NextGen App (Sovereign Trace)

6. Machuzu Secure Network & Cloud Node

Standard Encryption. Processing Through Node SSL Certificate. Premium services include: On-Chip Biometric Storage, Post Quantum Cryptography Encryption. Private Key Generation for non-eco-system blockchain based applications. Requires the Machuzu NextGen App.

What it can be used for

- To securely share and store/backup data within a decentralised network limited to participating Multiple Edge Nodes and a minimum 1 Cloud Node with Linux OS or Private Web Hosting Plan.

Integrations

- Machuzu ERP/HL7-FHIR EMR (eg: Supply Chain, FHIR-EMR data sharing)
- NextGen Blockchain (Identity, Sovereign Trace data sharing)
- Custom applications and API's.